

Abstract No.

Low grade glioma in Infancy and current neuro-oncology treatment standards in Germany

Isabel Fernandes Arroiteia¹, Helen Dzionsko¹, Julia Großer¹, Theresa M Thole-Kliesch¹, Connor Lindenmeyer², Olaf Witt^{3,4}, Till Milde⁵, Cornelis M. van Tilburg⁵, Brigitte Bison⁶, Lars Behrens⁷, Arend Koch⁸, David Capper^{9,10}, Ahmed El Damaty¹¹, Martin Ulrich Schuhmann¹², René Schmidt¹³, Moritz Fabian Danzer¹⁴, Amedeo M Azizi¹⁵, Svea Horn¹, Pablo Hernáiz Driever¹, and Ullrich-Wilhelm Thomale¹⁶

¹Charité-Universitätsmedizin Berlin, German HIT-LOGGIC-Registry for Low Grade Glioma in Children and Adolescents, Berlin, Germany

²Charité

³Hopp Children's Cancer Center Heidelberg (KiTZ), Heidelberg, Germany

⁴Clinical Cooperation Unit Pediatric Oncology, German Cancer Research Center (DKFZ), German Cancer Consortium (DKTK), Heidelberg, Germany

⁵Hopp Children's Cancer Center Heidelberg (KiTZ), German Cancer Research Center (DKFZ) and Heidelberg University Hospital, Heidelberg, Germany.

⁶8 Department of Neuroradiology, University Hospital of Augsburg, Augsburg, Germany

⁷University Hospital Augsburg

⁸12 Department of Neuropathology, Charité - Universitätsmedizin Berlin, Berlin, Germany

⁹Charité-Universitätsmedizin Berlin, Corporate Member of Freie Universität Berlin, Humboldt-Universität and Berlin Institute of Health, Berlin, Germany

¹⁰German Cancer Consortium (DKTK), German Cancer Research Center (DKFZ), Partner Site Berlin, Berlin, Germany

¹¹Department of Neurosurgery, Heidelberg University Hospital, Heidelberg, Germany

¹²Section of Pediatric Neurosurgery, Department of Neurosurgery, University Hospital Tübingen, Tübingen, Germany

¹³Institute of Biostatistics and Clinical Research, Münster, Germany.

¹⁴Institute of Biostatistics and Clinical Research, Münster, Germany

¹⁵Medical University of Vienna

¹⁶Charité-Universitätsmedizin Berlin, Department of Neurosurgery, Division of Pediatric Neurosurgery, Berlin, Germany.

Introduction: Low-grade gliomas (LGG) represent the most common type of solid brain tumor diagnosed in children and young adults in Germany. Since 2019, all confirmed cases have been centrally documented in a national registry. This registry supports the investigation of clinical and scientific questions by a multidisciplinary team. The present study aims to gain deeper insight into neurosurgical treatment patterns employed in infants diagnosed with low-grade glioma in Germany since the establishment of the registry.

Methods: This retrospective multicenter study is based on data collected by the national HIT-LOGGIC registry from 2019 to 2025. Infants aged 0 to 24 months at diagnosis with LGG were included. Collected data encompassed demographics, clinical characteristics, surgical interventions, neuropathological findings, and non-surgical treatments. The primary aim was to analyze neurosurgical management strategies within this cohort.

Results: A total of 174 patients were identified, with a median (Q1-Q3) age at diagnosis of 11.9 (5.6-



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18.8) months. Optic pathway was the most prominent localization, accounting for 47.7% of cases. The rate of primary tumor dissemination in this cohort was 15.4%. Pilocytic astrocytoma was diagnosed in 61% of cases. Surgical therapy alone achieved disease control in 29.9% of cases and was most frequently sufficient for hemispheric low-grade gliomas (75%) and cerebellar low-grade gliomas (68.4%). 53.4% of patients required additional systemic therapy. The overall neurosurgical adverse event rate considering all tumor resections or biopsies performed, was 28.1% (N=48/171) with clinical neurological events occurring in 18.1% (N=31/171) and surgery related complications in 14% (N=24/171), while 54.2% (N=26/48) were transient.

Conclusions: Infants with LGG represent a more challenging cohort of patients for the multidisciplinary team. Surgical treatment remains the most relevant initial therapy. While patient age alone does not significantly influence surgical decision-making, tumor location plays a critical role in determining the therapeutic efficiency.